

Machine Learning-Lecture 13

P1.py

```
C:\ml\L13>python p1.py
```

Day Weather Just Ate Will I Go Running?

0	1	Sunny	yes	yes
1	2	Rainy	yes	no
2	3	Sunny	no	yes
3	4	Rainy	no	no
4	5	Rainy	no	yes
5	6	Sunny	yes	yes
6	7	Rainy	no	no

Weather_Rainy Weather_Sunny Just Ate_no Just Ate_yes

0	False	True	False	True
1	True	False	False	True
2	False	True	True	False
3	True	False	True	False
4	True	False	True	False
5	False	True	False	True
6	True	False	True	False

Weather : 1 for Rainy and 2 for Sunny 1

Ate: 1 for no and 2 for yes 2

['no']

P2.py

```
C:\ml\L13>python p2.py
```

Day Weather Just Ate Will I Go Running?

0	1	Sunny	yes	yes
1	2	Rainy	yes	no
2	3	Sunny	no	yes
3	4	Rainy	no	no
4	5	Rainy	no	yes

5 6 Sunny yes yes

6 7 Rainy no no

Weather_Rainy Weather_Sunny Just Ate_no Just Ate_yes

0 False True False True

1 True False False True

2 False True True False

3 True False True False

4 True False True False

5 False True False True

6 True False True False

Weather : 1 for Rainy and 2 for Sunny 2

Ate: 1 for no and 2 for yes 1

['yes']

P3.py

C:\ml\L13>python p3.py

buying maintenance doors persons lug_boot safety class

0 vhigh vhigh 2 2 small med unacc

1 vhigh vhigh 2 2 small high unacc

2 vhigh vhigh 2 2 med low unacc

3 vhigh vhigh 2 2 med med unacc

4 vhigh vhigh 2 2 med high unacc

...

1722 low low 5more more med med good

1723 low low 5more more med high vgood

1724 low low 5more more big low unacc

1725 low low 5more more big med good

1726 low low 5more more big high vgood

[1727 rows x 7 columns]

buying 0

maintenance 0

doors 0

persons 0

lug_boot 0

safety 0

class 0

dtype: int64

	buying_high	buying_low	...	safety_low	safety_med
0	False	False	...	False	True
1	False	False	...	False	False
2	False	False	...	True	False
3	False	False	...	False	True
4	False	False	...	False	False
...	...	...	...	...	...
1722	False	True	...	False	True
1723	False	True	...	False	False
1724	False	True	...	True	False
1725	False	True	...	False	True
1726	False	True	...	False	False

[1727 rows x 21 columns]

class

unacc 1209

acc 384

good 69

vgood 65

Name: count, dtype: int64

	precision	recall	f1-score	support
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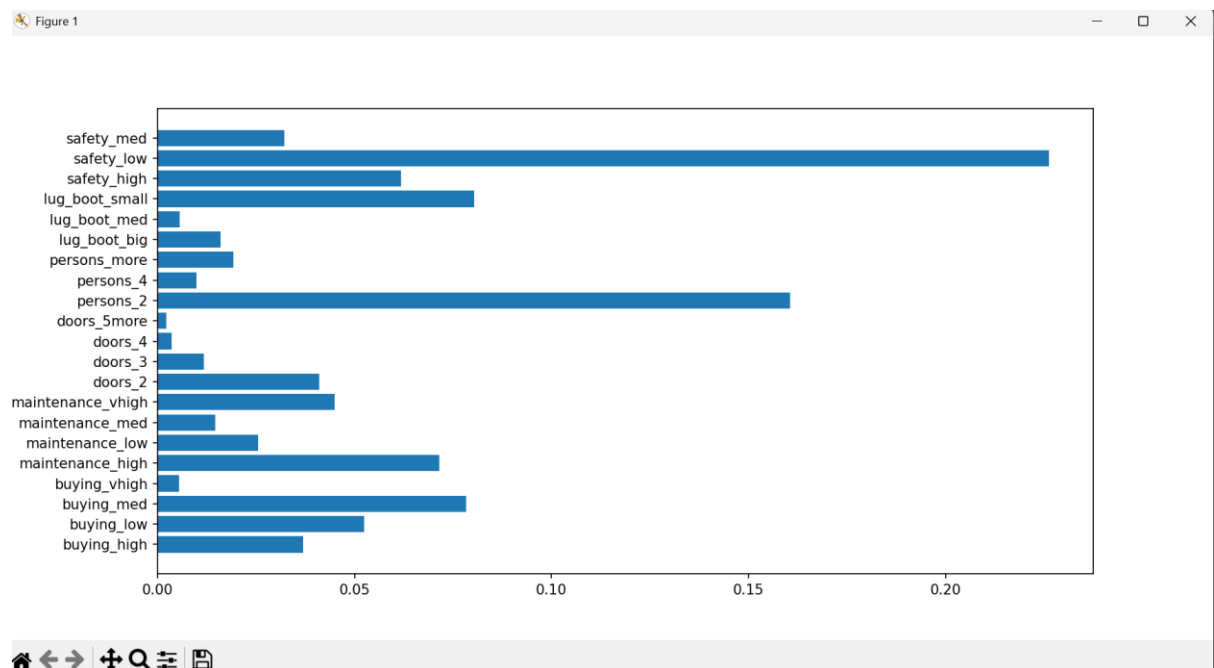
acc	0.92	1.00	0.96	109
-----	------	------	------	-----

good	1.00	0.95	0.97	20
------	------	------	------	----

unacc	1.00	0.98	0.99	287
vgood	1.00	0.81	0.90	16

accuracy		0.98	432	
macro avg	0.98	0.94	0.96	432
weighted avg	0.98	0.98	0.98	432

P4.py



P5.py

C:\ml\L13>python p5.py

```

buying maintenance doors persons lug_boot safety class
0  vhigh  vhigh  2  2  small  med unacc
1  vhigh  vhigh  2  2  small  high unacc
2  vhigh  vhigh  2  2  med  low unacc
3  vhigh  vhigh  2  2  med  med unacc
4  vhigh  vhigh  2  2  med  high unacc
...  ...  ...  ...  ...  ...  ...
1722 low  low 5more more med med good
1723 low  low 5more more med high vgood

```

1724	low	low	5more	more	big	low	unacc
1725	low	low	5more	more	big	med	good
1726	low	low	5more	more	big	high	vgood

[1727 rows x 7 columns]

persons 0

safety 0

class 0

dtype: int64

	persons_2	persons_4	...	safety_low	safety_med
0	True	False	...	False	True
1	True	False	...	False	False
2	True	False	...	True	False
3	True	False	...	False	True
4	True	False	...	False	False
...	...	...	...	...	...
1722	False	False	...	False	True
1723	False	False	...	False	False
1724	False	False	...	True	False
1725	False	False	...	False	True
1726	False	False	...	False	False

[1727 rows x 6 columns]

class

unacc 1209

acc 384

good 69

vgood 65

Name: count, dtype: int64

precision	recall	f1-score	support
-----------	--------	----------	---------

acc	0.55	1.00	0.71	105
good	0.00	0.00	0.00	15
unacc	1.00	0.82	0.90	293
vgood	0.00	0.00	0.00	19

accuracy			0.80	432
macro avg	0.39	0.45	0.40	432
weighted avg	0.81	0.80	0.78	432

P6.py

C:\ml\L13>python p6.py

```

    buying maintenance doors persons lug_boot safety class
0  vhigh   vhigh    2    2  small  med unacc
1  vhigh   vhigh    2    2  small  high unacc
2  vhigh   vhigh    2    2   med  low unacc
3  vhigh   vhigh    2    2   med  med unacc
4  vhigh   vhigh    2    2   med  high unacc
...  ...      ...    ...    ...    ...    ...
1722 low     low  5more  more   med  med  good
1723 low     low  5more  more   med  high vgood
1724 low     low  5more  more   big  low unacc
1725 low     low  5more  more   big  med  good
1726 low     low  5more  more   big  high vgood

```

[1727 rows x 7 columns]

buying 0

maintenance 0

doors 0

persons 0

lug_boot 0

safety 0

class 0

dtype: int64

	buying	maintenance	doors	persons	lug_boot	safety
0	vhigh	vhigh	2	2	small	med
1	vhigh	vhigh	2	2	small	high
2	vhigh	vhigh	2	2	med	low
3	vhigh	vhigh	2	2	med	med
4	vhigh	vhigh	2	2	med	high
...	...	...	...	...	...	...
1722	low	low	5more	more	med	med
1723	low	low	5more	more	med	high
1724	low	low	5more	more	big	low
1725	low	low	5more	more	big	med
1726	low	low	5more	more	big	high

[1727 rows x 6 columns]

	buying_high	buying_low	...	safety_low	safety_med
0	False	False	...	False	True
1	False	False	...	False	False
2	False	False	...	True	False
3	False	False	...	False	True
4	False	False	...	False	False
...	...	...	...	...	...
1722	False	True	...	False	True
1723	False	True	...	False	False
1724	False	True	...	True	False
1725	False	True	...	False	True
1726	False	True	...	False	False

[1727 rows x 21 columns]

[[-0.57757314 -0.57757314 -0.57757314 ... -0.70741389 -0.70649271

```

1.41359962]
[-0.57757314 -0.57757314 -0.57757314 ... 1.41359962 -0.70649271
-0.70741389]
[-0.57757314 -0.57757314 -0.57757314 ... -0.70741389 1.41544278
-0.70741389]
...
[-0.57757314 1.73138245 -0.57757314 ... -0.70741389 1.41544278
-0.70741389]
[-0.57757314 1.73138245 -0.57757314 ... -0.70741389 -0.70649271
1.41359962]
[-0.57757314 1.73138245 -0.57757314 ... 1.41359962 -0.70649271
-0.70741389]]
[[ 9.92467601e-01 -1.87518380e+00 -1.42577089e+00]
[ 9.92467601e-01 -1.87518380e+00 1.42577089e+00]
[-2.12019061e+00 7.80897469e-02 3.71283448e-01]
...
[-9.92467601e-01 1.87518380e+00 -6.51774309e-01]
[-1.48125322e-12 -2.43216510e-14 -2.07754520e+00]
[-2.87078460e-12 2.52795737e-13 7.73996579e-01]]
class
unacc 1209
acc 384
good 69
vgood 65
Name: count, dtype: int64

precision recall f1-score support

acc 0.53 0.54 0.54 94
good 0.38 0.18 0.24 17
unacc 0.85 0.90 0.88 295
vgood 0.39 0.27 0.32 26

```


accuracy			0.75	432
macro avg	0.54	0.47	0.49	432
weighted avg	0.74	0.75	0.74	432

P7.py

C:\ml\L13>python p7.py

```

    buying maintenance doors persons lug_boot safety class
0   vhigh    vhigh    2    2  small   med unacc
1   vhigh    vhigh    2    2  small  high unacc
2   vhigh    vhigh    2    2   med   low unacc
3   vhigh    vhigh    2    2   med   med unacc
4   vhigh    vhigh    2    2   med  high unacc
...   ...      ...    ...    ...    ...    ...
1722 low      low 5more  more   med   med good
1723 low      low 5more  more   med  high vgood
1724 low      low 5more  more   big   low unacc
1725 low      low 5more  more   big   med good
1726 low      low 5more  more   big  high vgood

```

[1727 rows x 7 columns]

buying 0

maintenance 0

doors 0

persons 0

lug_boot 0

safety 0

class 0

dtype: int64

```

    buying maintenance doors persons lug_boot safety
0   vhigh    vhigh    2    2  small   med

```

1	vhigh	vhigh	2	2	small	high
2	vhigh	vhigh	2	2	med	low
3	vhigh	vhigh	2	2	med	med
4	vhigh	vhigh	2	2	med	high
...	...	...	...	...	...	...
1722	low	low	5more	more	med	med
1723	low	low	5more	more	med	high
1724	low	low	5more	more	big	low
1725	low	low	5more	more	big	med
1726	low	low	5more	more	big	high

[1727 rows x 6 columns]

class

unacc 1209

acc 384

good 69

vgood 65

Name: count, dtype: int64

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

acc	0.50	0.52	0.50	99
good	0.11	0.06	0.08	17
unacc	0.84	0.86	0.85	303
vgood	0.27	0.23	0.25	13

accuracy		0.73	432
macro avg	0.43	0.42	0.42 432
weighted avg	0.72	0.73	0.72 432